Module 1

Quantum Mechanics

Contents:. Wave-Particle dualism, de-Broglie hypothesis, phase velocity, group velocity, relationship between phase velocity and group velocity, expression for de-Broglie wavelength from the concept of group velocity, Heisenberg's uncertainty principle, physical significance and application (broadening of spectral lines), Wave function Properties, Physical significance, derivation of Time independent Schrödinger wave equation, eigen value and eigen function, Application of Schrödinger Equation (one dimensional potential well) Numerical problems.

Extra Learning: Application of Schrödinger Equation (free particle) ,Quantum Tunelling

Introduction:

At the end of 19th century it was thought by the researchers and scientists that there were enough laws in physics using which any phenomena that occur in nature can be explained. All particles obey Newton's laws and all radiations obey Maxwell's equations. These were put together and are called Classical Physics.

During the same time, some experimental observations were made such as black body radiation, photo electric effect, Compton Effect. Using the existing theories and laws these observations could not be explained. In order to explain these new observations Quantum Physics was developed. These observations paved the way for the evolution of quantum mechanics which is put in to a broad category of Modern Physics. Any phenomena developed after this period is put under this broad umbrella. It deals with particles of atomic and subatomic dimensions. Before Quantum Mechanics, classical mechanics ruled the world of Physics. For particles down to atomic and subatomic dimension, classical mechanics was a failure and hence Quantum mechanics came into existence.

Wave Particle Dualism:

Dual nature of light:

In the beginning, light is considered to have wave nature and the phenomena such as reflection, refraction, interference, diffraction etc., can well be explained considering the wave nature of light.

The concept of black body radiation, photoelectric effect and Compton effect could not be explained based on the wave nature of light. To explain these observations it was assumed that light is made up tiny packets or particles called photons and energy can be absorbed or released only in terms of these photons. Hence it is concluded that light exists as both wave and particle nature depending on the circumstances. This paved the way for the dual nature of matter which was put forward by deBroglie.

Dual nature of matter & De Broglie hypothesis

Since nature loves symmetry, it was suggested by the scientist de Broglie that, when light which has predominantly wave nature can also exhibit the nature of particles under certain circumstances, we can assume that matter which has particle nature can also possess wave nature under suitable conditions. This assumption is called de Broglie hypothesis.

De Broglie suggested that, "When light can behave as particle then by all means, we can expect a particle to possess wave nature. The waves associated with matter are called matter waves or de Broglie waves, whose wavelength is given by $\lambda = \frac{h}{mv} = \frac{h}{p}$, where h is Planck's constant and p is linear momentum."

The waves associated with material particles of mass m moving with a velocity v are called matter waves. De-Broglie wavelength in terms of kinetic energy is,

$$\lambda = \frac{h}{\sqrt{2mE}}$$

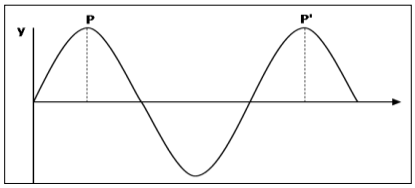
and for a charged particle of charge e, under the influence of the potential V, $\lambda = \frac{h}{\sqrt{2meV}}$

Characteristics of matter waves:

i) Matter waves are not electromagnetic in nature. They are called pilot waves.
ii) They are group waves travelling with a velocity V_g, same as the velocity of particle.
iii) Their wavelength is $\lambda = \frac{h}{p}$.
iv) They are complex in nature and has no physical significance.
v) They obey $V_p V_g = C^2$, where C is velocity of light in vacuum.
vi) The amplitude of matter waves depends on the probability of finding the particle.

Phase velocity V_p:

Each point in a travelling wave is represented by a unique phase. When a wave travels, the phase travels along with it. The velocity with which the wave or phase of the wave travels is called phase velocity V_p.



$$\boxed{V_p = \frac{\omega}{k}}$$

Group Velocity V_g:

When two or more waves travelling almost along same direction with same amplitude and slightly different frequency, wavelength are superposed, the resultant wave is enclosed inside wave packet. "The velocity with which the wave packet travels are called group velocity V_g".



Individual Waves



Amplitude variation after Superposition

$$\therefore \boxed{V_g = \frac{d\omega}{dk}}$$

Relation between Phase velocity and group velocity:

We know that,

$V_p = \frac{w}{k} \ldots\ldots\ldots\ldots (1)$

$V_g = \frac{dw}{dk} \ldots\ldots\ldots\ldots (2)$

From (1), $w = kV_p \ldots\ldots\ldots\ldots (3)$

Substituting (3) in (2), we get $V_g = \frac{d}{dk}(kV_p)$

$$= V_p + k\frac{dV_p}{dk}$$

$$V_g = V_p + k\,\frac{dV_p}{d\lambda}\cdot\frac{d\lambda}{dk} \ldots\ldots\ldots\ldots (4)$$

We know that,

$k = \frac{2\pi}{\lambda}; \quad dk = \frac{-2\pi}{\lambda^2}d\lambda$

$\therefore \frac{d\lambda}{dk} = \frac{-\lambda^2}{2\pi}$

$\therefore V_g = V_p + \frac{2\pi}{\lambda}\cdot\frac{dV_p}{d\lambda}\left(\frac{-\lambda^2}{2\pi}\right)$

$$\boldsymbol{V_g = V_p - \lambda\frac{dV_p}{d\lambda}}$$

Relation between Phase velocity and group velocity:

In terms of velocity of light:

$$\mathbf{V_g . V_p = C^2}$$

Relation between Particle velocity and group velocity:

In terms of velocity of light:

$$\mathbf{V_g = v}$$

Expression for de-Broglie wavelength $\lambda = \frac{h}{p}$

We know that the group velocity is given by,

$V_g = \frac{d\omega}{dk}$ ..... (1)

$w = 2\pi\vartheta; \quad d\omega = 2\pi d\vartheta$ ..... (2)

$k = \frac{2\pi}{\lambda}; \quad dk = 2\pi d(1/\lambda)$ ..... (3)

$\therefore \quad V_g = \frac{2\pi dv}{2\pi d\left(1/\lambda\right)} = \frac{dv}{d\left(\frac{1}{\lambda}\right)}$ ..... (4)

We know that, $E = \frac{1}{2}mv^2 + V$

Also, $E = h\vartheta$

$\therefore \quad h\vartheta = \frac{1}{2}mv^2 + V$ ..... (5)

Differentiating (5), $hd\vartheta = \frac{2mv\,dv}{2}; \quad d\vartheta = \frac{mv}{h}dv$ ..... (6)

Subtracting (6) in (4) $V_g = \frac{mv\,dv}{hd\left(1/\lambda\right)} \quad (\because V_g = v)$

$d\left(\frac{1}{\lambda}\right) = \frac{m}{h}dv$ ..... (7)

Integrating (7) assuming integration constant = 0

We get, $\frac{1}{\lambda} = \frac{mv}{h}; \quad \boldsymbol{\lambda = \frac{h}{mv} = \frac{h}{p}}$

Heisenberg's Uncertainty Principle (HUP):

According to classical mechanics a particle occupies a definite place in space and possesses a definite momentum. If the position and momentum of a particle is known at any instant of time, it is possible to calculate its position and momentum at any later instant of time. The path of the particle could be traced. This concept breaks down in quantum mechanics leading to **Heisenberg's Uncertainty Principle** according to which "there are certain pair of physical quantities which cannot be measured simultaneously and accurately. The product of the uncertainties involved in their measurement is greater than or equal to $\frac{h}{4\pi}$. For example, it is impossible to measure simultaneously both the position and momentum of a particle accurately. If we try to measure very accurately the position of a particle, it leads to large uncertainty in the measurement of momentum and vice versa.

If Δx and ΔP_x are the uncertainties in the measurement of position and momentum of the particle then the uncertainty can be written as

$$\Delta x \,.\, \Delta P_x \geq \mathrm{h}/4\pi$$

$$\Delta E . \Delta t \geq \frac{h}{4\pi}$$

$$\Delta L . \Delta \theta \geq \frac{h}{4\pi}$$

Where,

$\Delta x \rightarrow uncertainity\ in\ position,\ \ \Delta P \rightarrow uncertainity\ in\ momentum$

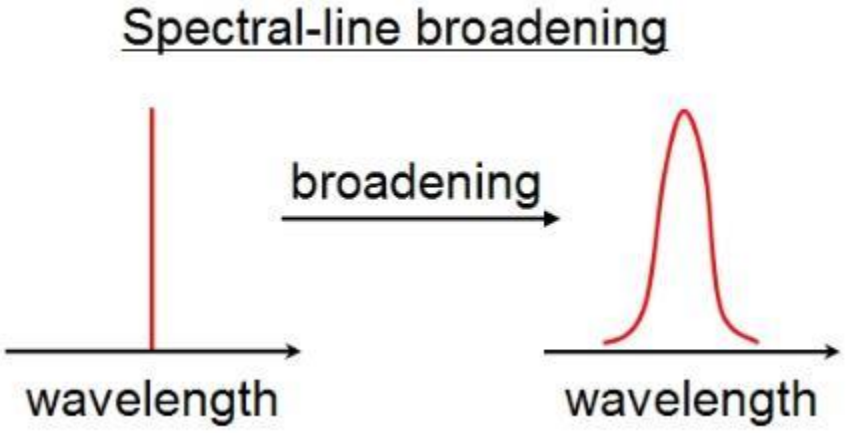
$\Delta E \rightarrow uncertainity\ in\ energy,\ \ \Delta t \rightarrow uncertainity\ in\ time$

$\Delta L \rightarrow uncertainity\ in\ angular\ momentum,\ \ \ \ \Delta\theta \rightarrow uncertainity\ in\ angular\ position$

$h \rightarrow Planck's\ constant$

Application (or) example:

1. **Broadening of spectral lines:**
   An individual atom/molecule making a transition between energy levels emits one photon with a well-defined energy / frequency. However, profiles of real spectral lines are not infinitely narrow. The broadening in the spectral lines is observed due to the indeterminateness in the atomic energies.



Using the HUP,

$$\Delta E \Delta t \geq \frac{h}{4\pi}$$

$$\Delta E \geq \frac{h}{4\pi * \Delta t}$$

Where,

ΔE is the uncertainty in the measurement of energy

Δt is the mean life time of a level

Δt is finite and of the order of 10^{-8} sec. Therefore ΔE also has a finite energy spread, which means the energy levels are not sharp and hence the broadening of the spectral lines. For the energy to be sharp, the time should be infinite which is not possible and hence there is a natural broadening of the spectral lines.

Significance of HUP :

i) It signifies dual nature of matter.

ii) From $\Delta x \Delta p \geq \frac{h}{4\pi}$, it is clear that if Δx decreases then Δp increases. If we try to measure x accurately, the error in measuring p increases, which will lead to the measurement of absurd values. Hence it is always proper to measure the **most probable value** of physical quantity than accurate values.

Wave function (ψ) : Wave function in quantum mechanics is a variable quantity that mathematically describes the wave characteristics of a particle. The value of wave function of a particle at a given point of space and time is related to the likelihood of the particle being there at that time. It is a function of position and time, which gives information about a particle completely. Usually wave functions are complex in nature. Wave function is related the probability of finding the particle at a given place and given time. It is represented by "psi – ψ"

Schrodinger's equation :

The basic equation in quantum mechanics is given by Schrodinger which is of two types: Time dependent and time independent. By solving the equation we can determine the Eigen function and Eigen values of a particle.

The complex wave equation can be represented as,

$$\psi = Ae^{-iw\left(t-\frac{x}{v}\right)} \ldots\ldots\ldots (1)$$

Differentiating (1) partially w.r.t x twice,

$$\frac{\partial \psi}{\partial x} = Ae^{-iw\left(t-\frac{x}{v}\right)} * \frac{iw}{v}$$

$$\frac{\partial^2 \psi}{\partial x^2} = Ae^{-iw\left(t-\frac{x}{v}\right)} * \frac{i^2 w^2}{v^2}$$

$$\frac{\partial^2 \psi}{\partial x^2} = -4\pi^2 \frac{\gamma^2}{v^2}\psi \qquad \left[\because\ i^2 = -1,\ w^2 = (2\pi\gamma)^2 = 4\pi^2\gamma^2, Ae^{-iw\left(t-\frac{x}{v}\right)} = \psi\right]$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2}{v^2}\psi \qquad \left[\because \frac{\gamma^2}{v^2} = \frac{1}{\lambda^2}\right]$$

$$\therefore \frac{1}{\lambda^2} = \frac{1}{4\pi^2\psi} \cdot \frac{\partial^2\psi}{\partial x^2} \ldots\ldots\ldots (2)$$

We know that, the total energy $E = \frac{p^2}{2m} + V$

$$(E - V) = \frac{h^2}{2m\lambda^2} \quad \left[\because p = \frac{h}{\lambda}\right]$$

$= \frac{h^2}{2m}\left[-\frac{h^2}{4\pi^2\psi}.\frac{\partial^2\psi}{\partial x^2}\right]$ (Substituting for $\frac{1}{\lambda^2}$ from (2))

$$= \frac{-h^2}{8\pi^2 m\psi}.\frac{\partial^2\psi}{\partial x^2}$$

$$\therefore \frac{8\pi^2 m}{h^2}(E - V)\psi = -\frac{\partial^2\psi}{\partial x^2}$$

$\boxed{\frac{\partial^2\psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0}$ is Schrodinger's time independent equation in one dimension.

In 3 dimension. $\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

Eigen function and Eigen values; significance of ψ (physical interpretation of ψ ; Max-Born interpretation; Normalisation condition):

We know that the Schrodinger equation is the 2nd order partial differential equation in ψ and is,

$$\frac{\partial^2\psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

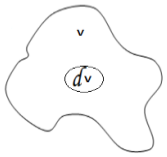
The solution of the above equation gives ψ, which has more that 1 value. The solutions have to satisfy the following conditions, for them to be called as eigen function.

If the solutions are,

i) Single valued w.r.t x
ii) Continuous w.r.t to x
iii) Finite and
iv) For bound states 'ψ' must vanish at potential boundary and outside. If 'ψ*' is a complex function, then ψ* ψ must also vanish at potential boundary and outside, then they are called Eigen function / wave function.

Therefore the wave function should be finite, continuous and single valued.
For each eigen function we get corresponding energy values and they are called eigen values.

Significance :



ψ has no physical significance. It is $|\psi^2|$ which has the significance. $|\psi^2|$ is called probability density function.It is the probability of finding a particle in unit volume.

The probability of finding a particle per unit volume = $|\psi|^2$.

Let us consider a particle in a volume V, which is divided into elementary volumes dV.

∴ Probability of finding the particle in volume dV = $|\psi|^2 dV$

The probability of finding the particle in volume V = $\int_V |\psi|^2 dV$

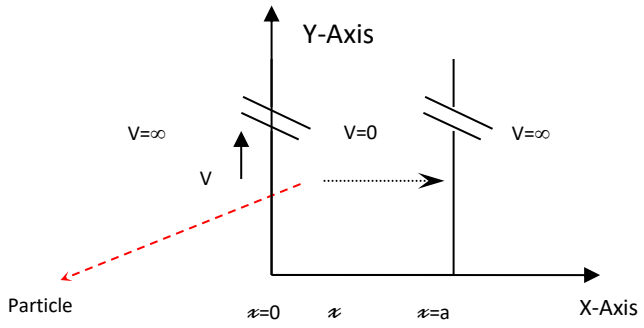
Since, the particle is present in the volume, probability of finding it is 1.

$$\therefore \int_V |\psi|^2 dV = 1$$

This is called normalization condition, Max-Born interpretation or significance of ψ.

Application :

Particle in a one dimension potential well of infinite height (or) particle in a box.



Consider a potential well of width 'a' and height ∞. Particle is inside the well where V=0. Outside and on the walls of the well V = ∞. We are going to find the Eigen function and Eigen values for the particle.

We know that, $\frac{\partial^2\psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$ ... ... (1)

Outside the well V=∞

$$\therefore \frac{\partial^2\psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E - \infty)\psi = 0 \text{ ... ... (2)}$$

The above equation is true only for $\psi = 0$

$$\therefore \psi = 0 \; at \; x = 0 \; \& \; x = a \text{ ... ...(3)}$$

Inside the well V = 0

$$\therefore \frac{\partial^2\psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}E\psi = 0$$

$$Let \; \frac{8\pi^2 m}{h^2}E = k^2 \text{ ... ... (4)}$$

$$\frac{\partial^2\psi}{\partial x^2} + k^2\psi = 0 \text{ ... ...(5)}$$

The solution of equation (5) is

$$\psi = A\, coskx + B \sin kx \text{ ... ...(6)}$$

$$At \; x = 0, \psi = 0$$

$$\therefore A = 0 \text{ ... ...(7)}$$

Hence $\psi = B\, sinkx$ ... ...(8)

At x = a, $\psi = 0$

$\therefore B \sin ka = 0$ (B cannot be zero)
$\therefore \sin ka = 0$
$ka = n\pi; \quad k = \frac{n\pi}{a}$ ... ...(9)

Substitute k in equation (4)

$$\frac{8\pi^2 m}{h^2}E = \frac{n^2\pi^2}{a^2}$$

$\therefore E = \frac{n^2h^2}{8ma^2}$ ... ...(10) n = 1,2,3,…….

[n = 0 not allowed as $\psi = 0$ at n =0]

The value of energy for n = 1 is called zero point energy or ground state energy.

$\therefore \quad E_z = \dfrac{h^2}{8ma^2}$

The values of energy for n = 2, 3, 4 …….. are called excited states.

$E_1 = 4E_z$, $E_2 = 9E_z$ and so on.

$\therefore$ The Energy of the particle is quantized.

To find B : Using normalization condition for 1 dimension

$$\int_{x=0}^{a} |\psi|^2 \, dx = 1$$

$$\int_{o}^{a} B^2 \sin^2 kx \, dx = 1; \qquad \int_{0}^{a} \frac{B^2}{2}(1 - cos2kx)dx = 1$$

To find B : We know that using normalization technique,

$$\int_{x=0}^{x=a} |\psi|^2 \, dx = 1$$

$$\int_{o}^{a} B^2 \sin^2 \frac{n\pi}{a} x \, dx = 1$$

$$B^2 \left[\int_0^a \frac{dx}{2} - \frac{1}{2} \int_0^a cos \frac{2n\pi x}{a} dx \right] = 1$$

$$\therefore \quad \frac{B^2 a}{2} = 1; \quad B^2 = \frac{2}{a}; \quad B = \sqrt{\frac{2}{a}}$$

$\therefore \quad \psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x \quad$ & $\quad E_n = \frac{n^2 h^2}{8ma^2}$ are the values of ψ & E for a particle in one

Dimension potential well.

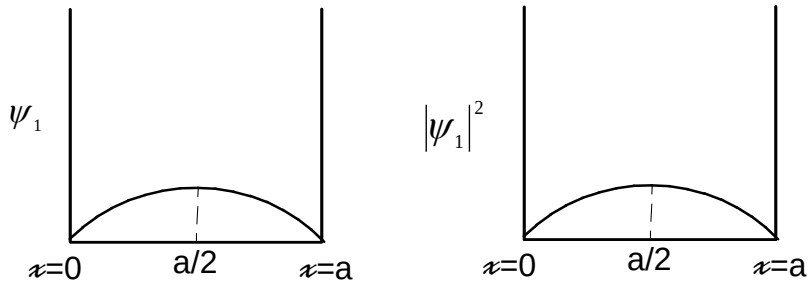
Case i) : n = 1 : Here, $E = E_z = \frac{h^2}{8ma^2}$ and $\psi = \sqrt{\frac{2}{a}} \sin \frac{\pi}{a} x$

At x = 0 & x = a, $\psi = 0$. $\therefore |\psi|^2 = 0$

Hence the probability of locating the particle at walls = 0

$\psi = \sqrt{\frac{2}{a}}$ when x = $\frac{a}{2}$ $\quad \therefore |\psi|^2 = \frac{2}{a}$ ($maximum$)

Hence the particle is located mostly at the centre of the well.



Case ii) : n = 2

$$E = 4E_z = \frac{4h^2}{8ma^2}$$

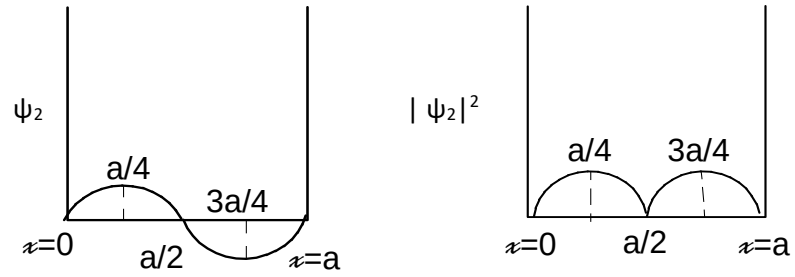
$$\psi = \sqrt{\frac{2}{a}} \sin \frac{2\pi}{a} x$$

At x = 0 & x = a, $x = \frac{a}{2}$, $\psi = 0$. $\therefore |\psi|^2 = 0$

$\therefore$ The probability of locating the particle at the walls and at the center of the well = 0

At $x = \frac{a}{4}, \frac{3a}{4}$; $\psi = \sqrt{\frac{2}{a}}$ (or) $|\psi|^2 = \frac{2}{a}$ (maximum)

$\therefore$ The probability of locating the particle at ¼ th & 3/4th of well width is maximum.



Case iii) : n = 3

$$E = 9E_z = \frac{9h^2}{8ma^2}$$

$$\psi = \sqrt{\frac{2}{a}} \sin \frac{3\pi}{a} x$$

At x = 0 & x = a, $x = \frac{a}{3}, \frac{2a}{3}$; $\psi = 0$. $\therefore |\psi|^2 = 0$

At $x = \frac{a}{6}, \frac{3a}{6}, \frac{5a}{6}$; $\psi = \sqrt{\frac{2}{a}}$ (or) $|\psi|^2 = \frac{2}{a}$ (maximum)

